

Sai Sneha Kakarla

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SUMMARY

Software Engineering master's student experienced in building distributed, microservices-based backends and integrating large language models (LLMs) into production-style applications. Comfortable across the full development lifecycle - design, coding, testing, and deployment and quick to adapt in ambiguous environments. Brings 3.5 years of government IT experience in administrating and maintaining enterprise systems and databases..

TECHNICAL SKILLS

- **Languages:** Python, C++, C, SQL, JavaScript
- **Distributed Systems & Backend:** Flask (REST APIs, microservices), Apache Kafka (event streaming), Nginx API Gateway, Server-Sent Events (SSE), PostgreSQL, MongoDB, MySQL
- **AI & LLMs:** Model Context Protocol (MCP), Groq LLM API, classification trees, decision tables, security testing
- **DevOps & Tools:** Docker, Docker Compose, Git/GitHub, CI/CD, Postman, Appium, Selenium
- **Frontend:** React.js, HTML/CSS, JavaScript
- **Concepts:** Microservices, distributed systems, system design, CAP theorem, Agile, TOGAF

PROJECTS

Smart Hotel Operations Platform | CMPE 272, SJSU Group Project

Jan – May 2026

github.com/snehak9721/smart-hotel-platform

- Architected an end-to-end microservices backend (Flask, Apache Kafka, PostgreSQL, MongoDB, Docker Compose) designed for high-throughput, asynchronous data flow.
- Implemented Kafka event publishing and consumption for real-time room-status streaming across services, enabling scalable event-driven communication.
- Built an Nginx API Gateway routing requests across 3 Flask microservices, with SSE streaming endpoints powering live occupancy dashboards.
- Integrated an LLM-powered chat service using the Model Context Protocol (MCP) and Groq API, connecting a React.js frontend to backend REST APIs.
- Orchestrated 7 Docker containers (Zookeeper, Kafka, PostgreSQL, MongoDB, 3 Flask services) via Docker Compose; authored architecture diagrams and full GitHub documentation.

AI Food Recognition — Quality Assurance & Test Automation | CMPE 287, SJSU Group Project Jan – May 2026

github.com/snehak9721/Nutri-Snap-Test-Automation-using-Appium

- Built Python automated test scripts (Appium) for UI and API validation, executed on a real mobile device.
- Designed and executed test suites using AI-based classification trees, 3D decision tables, and conventional methods aligned with the AI Testing framework.
- Performed security testing and authored formal test-design documents with full traceability to requirements, collaborating within a QA team.

Automatic Segmentation and Recognition of Lung Region | B.Tech Major Project (MATLAB)

2017 – 2018

- Developed a MATLAB image-processing pipeline to segment and analyze CT-scan lung images, applying detection algorithms to localize potential tumor areas.

WORK EXPERIENCE

Digital Assistant & Nodal IT Officer - AP State Government, India

2019 – 2023

- Managed government web portals and enterprise databases, ensuring data accuracy, consistency, and timely availability across operations.
- Administered IT infrastructure — mobile applications, tablets, desktop systems, and network connectivity - serving as nodal officer for all IT-related activities.
- Monitored and administered government data within enterprise management suites to maintain data integrity and reporting compliance.

EDUCATION

M.S. in Software Engineering - San Jose State University, CA

2025 – 2027 (Expected)

B.Tech in Electronics & Communication Engineering - JNTU, India

2014 – 2018